

## NON-LETHAL WEAPONS IN THE NEWS

Defense Electronics February, 1993 p. 41

ISSN: 0278-3479

This is a summary of a rather long article:

DOD: There is strong support for taking the effort for development of non-lethal weapons away from the individual services and making an independent office. Among those who support such a move are: Chris and Janet Morris of the Washington DC based United States Global Strategy Council (GSC). These are lobbyists who have been pushing to have the control placed under the OSD. Subsequently, Dick Cheney appointed a "Non-lethal weapons study group. The detractors of this effort are Deputy Secretary of Defense Donald Atwood, Under Secretary of Defense for Acquisitions Donald Yockey, and Dr Victor Reis, former director of the Defense Advanced Research Projects Agency (DARPA) and the Defense, Research and Engineering Directorate. Their reason for not supporting such a move is they do not want any congressional oversight and are afraid if there is, congress will use it to gut the DoD budget. Los Alamos National Laboratory (LANL): John Alexander, disabling technologies program director. "It is postulated that the development of non-lethal weapons systems would be a deterrent in some situations. For example, currently, drug smugglers are 100 percent convinced the the US will never employ lethal force against them if they do not shoot first. Many air interdiction have ended with the smugglers turning tail and running, confident that they will not be shot down. A non-lethal means of controlling flight would serve as a deterrent in that situation."

ARMY: Major effort in this area is being conducted at the Picatinny Arsenal in New Jersey, Armament Research Development and Engineering Center (ARDEC) under a program called "Low Collateral Damage Munitions (LCDM). Training and Doctrine Command (TRADOC): A 9/4/92 draft report entitled 'Operations Concept for Disabling Measures' was prepared by the Army's Training and Doctrine Command, Fort Monmouth, NJ and is currently circulating. The draft report describes the applicability of non-lethal capabilities in providing civilian policy-makers and military commanders a new range of options for containing a crisis before it reaches a lethal plateau. A formal framework for studying the use of disabling measures vs the use of lethal force is provided by the report. Disabling or non-lethal measures would include the use of infrasound devices to cause disorientation among troops, machinery clogging chemical compounds, lasers to blind sensors, and microwave weapons to destroy electronics. The draft document is considered to be the initial step in the development and purchase of such weapons systems. The article contains additional information on the aforementioned effort.

Programs:

Optical Munitions, disrupt enemy sensors.

High Power Microwave (HPM) Projectiles, knock out command and control.

Acoustic Beams, [Scientific Applications & Research Associates (SARA) of Huntington Beach, California - contractor/consultant], An acoustic "bullet" would incapacitate by creating "plasma in front of the target, which creates an impact wave that is just like a blunt object. It causes blunt object trauma, like being hit by a baseball."

Pulsed Chemical LASERS, "... create a hot, high pressure plasma in the air in front of a target surface, creating a blast wave that will result in variable, but controlled effects on materiel and personnel."

LASER Dye Rods, "exceptionally intense flash, blinding surrounding optical sensors and personnel"

\*\*\*\*\*

Defense News April 15, 1991 p. 8  
ISSN: 0884-139X

The US Army Materiel Command's Picatinny Arsenal (New Jersey) wants to develop infantry weapons that would use laser, sound, radio, or particle beams to allow selective incapacitation or killing of hostile troops and equipment. The Joint Service Small Arms Program Management Committee is sponsoring the program, according to Matt Zimmerman, project dir. The effort would comprise two phases, one 10 mo long to study possible weapons, and one 18 mo long to demonstrate them in the laboratory, he says; firms may be paid to take part, he says. Such non-lethal weapons, Zimmerman says, could be used to minimize collateral damage.

ARDEC is focusing on arms ranging from the 9 mm pistol to the 8 in (203 mm) howitzer, work is also underway in other parts of the army in developing modular warheads (which could include a family of non-lethal munitions) on other types of launch systems.

\*\*\*\*\*

Aerospace Daily August 24, 1992; Pg 312; Vol. 163, No. 38  
Journal Code: ASD ISSN: 0193-4546  
Section Heading: What's Ahead In Aerospace  
Word Count: 100

TEXT:

Senate Armed Services Committee Chairman Sam Nunn (D-Ga.) links the idea of a U.S. peacemaking or peacekeeping force with that of non-lethal technology. "I think any peacekeeping force has got to have an ability to defend itself with lethal weapons," Nunn tells reporters, "but there are non-lethal type capabilities that are within our reach that ought to be looked at very carefully and exploited." Since the Persian Gulf war, the Pentagon has been trying to determine the proper emphasis to give such technology,

although officials there prefer the term "disabling." They say "non-lethal weapon" is an oxymoron (DAILY, March 6).

Copyright 1992 McGraw-Hill, Inc.

\*\*\*\*\*

01757569/9

01757569 DIALOG FILE 636: PTS NEWSLETTER DATABASE

NON-LETHAL WEAPONS OFFER NEW SO/LIC CAPABILITIES

Tactical Technology February 03, 1993 V. 3 NO. 3 p. 1

WORD COUNT: 1611

PUBLISHER: Phillips Business Information, Inc.

Prior to Desert Storm, when the Iraqis held 1,000 U.S. citizens captive, one option explored for their rescue involved air-dropping infrasound transmitters to disable and disorient the guards.

Infrasound transmitters could have been dropped in the area, causing the Iraqi guards and their prisoners to become extremely sick while special operations troops performed the rescue. Infrasound, or very low frequency transmitters cause disorientation, nausea, vomiting and/or bowel spasms. It would have created a form of seasickness on the ground. A major drawback of the plan was that the hostages would have been equally effected by the transmissions, requiring about 3,000 special operations forces to rescue the 1,000 hostages. Fortunately the Iraqis released the hostages and neither this nor any other rescue plan was required.

Although the use of infrasound in this situation was not feasible, it and many other non-lethal technologies offer a wide variety of options for both military and law enforcement applications says Janet Morris, research director at the U.S. Global Strategy Council, a Washington, D.C.-based think-tank founded by Ray Cline, a former deputy director of the CIA. Morris is one of the strongest proponents for the establishment of an office of non-lethality in the Clinton administration.

The U.S. has entered a new age of accountability. Both the military and law enforcement are held accountable for their actions. Fratricide, killing of civilian bystanders, destruction of civilian property are all liability issues that authorities are facing. Many police departments have suspended high-speed car chases for fear of innocent bystanders being hurt and the resulting law suits. Imagine police armed with an electronic gun that could disable the electronic ignition of a car, say proponents of non-lethal weapons.

"Think of a policeman rounding a corner to be confronted by a boy with an Uzi pointing at the officer. The officer can't tell if it's a real Uzi or a toy gun. Presently the officer has a limited number of choices. What if he had a non-lethal device that he could use to disable the boy, without permanent damage, and find out afterwards if the Uzi is real or not," Morris said during a telephone interview with Tactical Technology. Offering police an effective

weapon that will disable without causing permanent damage will give officers greater flexibility. U.S. troops in Somalia face similar dilemmas. With the number of low-intensity conflicts (LIC) around the world increasing on a daily basis, the U.S. will likely find itself involved. Although no office of non-lethality exists today, the U.S. military and other agencies have launched research into this area.

**If the Clinton administration creates an office of non-lethality, Morris envisions it as being responsible for developing:**

- \* **a non-lethality policy statement,**
- \* **a non-lethality National Security Decision, and**
- \* **a non-lethal strategic doctrine.**

In addition, Morris envisions the office helping to define terms and set policies for U.S. involvement in peacemaking, peacekeeping, and support to humanitarian actions below the threshold of war. During the interview, Morris said there is growing pressure on the President to create such an office. An exciting aspect of non-lethality is that many of the needed technologies either already exist or are being developed. However, many of the technologies require additional work in making them deployable in a tactical situation.

The U.S. is not the only country interested in non-lethal weaponry. According to Morris, the military attache at the Russian Embassy has contacted her organization about the possibility of converting Russian military hardware to a non-lethal capability. She also said France is exploring the use of infrasound for crowd control. The closer you are to the infrasound transmitter, the worse you feel. By placing infrasound transmitters in front of a rioting crowd, the crowd will self-disperse.

Morris says that the ability to control minds and probe psyches exists in the basement of a mental hospital in Moscow.

While visiting Russia in November 1991, Morris and other members of a team sent to investigate Russian technologies for commercial development were invited to a demonstration of a mind control technology. A volunteer from the U.S. team sat down in front of a computer screen as innocuous words flashed across the screen. The volunteer was only required to tell which words he liked and which words he disliked. At the end of the demonstration the Russian staff started revealing the sensitive, innermost thoughts of the volunteer--none of which had been previously discussed.

A microphone was presented to Morris and she was told to speak into it. She declined. A Russian took the microphone back, grinned and said into the microphone, "Vote for Yeltsin."



The recorded message was mixed with what appeared to be white noise or static, so when played back it became undecipherable. Since there were no more volunteers in the U.S. group, the Russians volunteered to go upstairs and let the Americans choose a mental patient for a demonstration. The Americans declined the offer.

The Russians told Morris of a demonstration in which a group of workers were outside the hospital working on the grounds. The staff sent an acoustic psycho-correction message via their machine to the workers telling them to put down their tools, knock on the door of the hospital and ask if there was anything else they could do. The workers did exactly that, the Russians said. The Russians admitted to using this technology for special operations team selection and performance enhancement and to aid their Olympic athletes and an Antarctic exploration team. Unlike lie detectors, this machine can determine when the truth is spoken, according to Morris.

Being an infrasound, very low frequency-type transmission, the acoustic psycho-correction message is transmitted via bone conduction. This means that earplugs will not restrict the message. An entire body protection system would be required to stop reception. The message, according to the Russians, bypasses the conscious level and is acted on almost immediately. The Russians say that the messages are acted upon with exposure times of under one minute.

Morris envisions this technology will be miniaturized into a hand-held device. Presently, the International Healthline Corp. of Richmond, Va., is planning to bring a Russian team of specialists to the U.S. in the near future to further demonstrate the capability. International Healthline is a private corporation that is exploring Russian medical technologies for import to the U.S.

"This is just one of hundreds upon hundreds of exceedingly advanced medical technologies that exist today in Russia," said Jim Hovis, International Healthline's CEO. "There are advanced medicines for the treatment of wounds, burns, radiation and a whole host of other medical challenges."

Healthline hopes to screen, obtain and develop Russian medications that fall into the \$3 to \$5 billion annual sales category market. Hovis sees the acoustic psycho-correction device as having military and civilian medical applications.

"In addition to performance enhancement, this device could also be used as a substitute for present anesthetics," Hovis said.

The device was pioneered by the Russian Department of Psycho-Correction at the Moscow Medical Academy. Although many spin-off applications of this technology are possible, it was originally developed to aid mentally disturbed individuals.

The thought of such a device leads to a plethora of scary possibilities. When should such a device be used? Who should be given access to the device?

And what controls will be needed to avoid abuse? How can one detect if one has been the subject of mind control abuse? The Russian official in Morris' story said, "Vote for Yeltsin." One wonders why he picked that suggestion.

#### TABLE--NON-LETHAL TECHNOLOGIES

- Infrasound:** Very low frequency sound generators can be tuned to incapacitate humans, causing disorientation, nausea, vomiting, or bowel movements. By adjusting the signal, infrasound can also be used to crumble masonry.
- Laser Weapons:** Low-energy laser rifles with power packs can flash-blind people and disable optical and infrared systems used for target acquisition, tracking, night vision and range-finding.
- Isotropic Radiators:** Modified conventional weapons can fire omnidirectional laser-bright rounds that will dazzle people or optics.
- Non-Nuclear Electromagnetic Pulse (EMP):** Developmental beam generators producing one gigawatt of power could be used to explode ammunition dumps within line of sight or paralyze unprotected in-use electronic systems, including systems with antennas or systems connected to batteries. Vulnerable systems include all engines with electronic ignition systems, gasoline pumps, radars, communications, navigation, and unshielded electronic triggers of explosive and nuclear devices.
- High Power Microwave (HPM):** A national program exists for the development of HPM systems designed to disrupt electronics. HPM pulses can either be fired repetitively or HPM bombs can be made to disable electronic devices.
- Liquid Metal Embrittlement (LME):** LME agents work by changing the molecular structure of base metals or alloys to which it is applied. LMEs are clear and have little or no perceptible residue, whether sprayed on or applied with felt-type markers. Used selectively on critical metal structures--aircraft, ships, trucks, elevators, metal treads, bridge supports--LMEs can cause a significant disruption and psychological distress.
- Light Detection and Ranging (LIDAR):** LIDAR can be used for plum identification and sensor blinding. LIDAR systems could be used to identify the plume of covert drug manufacture.
- Supercaustics:** Based on superacid technology, supercaustics can be millions of times more caustic than hydrofluoric acid. A round that delivers jellied superacids could destroy optical systems, glass blocks, or vision blocks.
- Calmative Agents:** Calmatives, or sleep agents, mixed with DMSO (quickly delivers chemicals through the skin into the bloodstream) could be used in riot control, counterinsurgency, or even in selected hostage situations.

**Anti-Traction Technology:** Anti-traction lubricants delivered by aircraft or human agents can render railroads, ramps, runways and stairs ineffective for limited periods of time.

**Polymer Agents:** Polymers in burst munitions could foul air-breathing engines. Polymer adhesives can also effectively glue equipment in place or keep it from operating.

**Combustion Alteration:** Internal combustion engines can be disrupted through special munitions. These munitions would temporarily contaminate fuel or change its viscosity to degrade engine performance.

Source: Janet Morris, U.S. Global Strategy Council

\*\*\*\*\*

01757568      DIALOG FILE 636: PTS NEWSLETTER DATABASE  
ARMY EXPLORING LESS THAN LETHAL TECHNOLOGIES  
Tactical Technology    February 03, 1993    V. 3    NO. 3  
WORD COUNT: 838  
PUBLISHER: Phillips Business Information, Inc.

Challenged by the Defense Department, the Army's Armament Research, Development & Engineering Center (ARDEC) has launched a series of research programs that can effectively render the enemy incapable while minimizing collateral damage.

Through the Low Collateral Damage Munitions (LCDM) Programs, the Army hopes to develop technologies that can disable, dazzle or incapacitate aircraft, missiles, armored vehicles, personnel and other equipment without causing extensive collateral damage. In general, the Army hopes the program payoffs will include:

- \* Increased tactical flexibility in future conflicts;
- \* Dramatic performance enhancements beyond those obtainable from conventional bullets and bursting munitions;
- \* Low collateral damage capability/minimization of loss of life;
- \* Increased hit/incapacitation probability through line-of-sight projection, eliminating current ballistic trajectory;
- \* Improved logistics/soldier load with bullet-less systems.

The LCDM program will explore non-conventional means of incapacitating unarmored/armored materiel and protected/ unprotected personnel. ARDEC,